CDI COLLEGE

Assignment 2

Technical Analysis Using R – Preliminary Stage

Course	BDA400 – Data Science Tools and Techniques
Student	Mohamed Elahwel
Assignment	CA_BDA400_A02
Portfolio symbols	AAPL, MSFT, NVDA
Required filename	Elahwel_Mohamed_CA_BDA400_A02.pdf
GitHub repository	github.com/mohamedelahwelca-hub/TechnicalAnalysis

Project scope: configure an R workflow, read the selected stock symbols, retrieve historical price data, calculate descriptive statistics, create closing-price charts, and save a CSV summary.

Evidence note: this report includes genuine screenshots supplied from the student's RStudio and GitHub sessions. The screenshots demonstrate project execution and repository contents; they do not document the original R/RStudio installation process.

1. Introduction and Objectives

This preliminary stage establishes the technical environment for a three-stock technical-analysis project. The selected symbols are Apple (AAPL), Microsoft (MSFT), and NVIDIA (NVDA). The workflow uses R and RStudio to import time-series stock data, calculate summary measures, display recent observations, and generate visualizations.

Project objectives

- Organize the project in a GitHub repository named TechnicalAnalysis.
- Configure an RStudio working directory and required R packages.
- Read one ticker symbol per line from portfolio.txt.
- Import historical market data and calculate basic statistics.
- Create closing-price charts and export a CSV statistics file.

2. GitHub Repository and Project Files

The public GitHub repository is named TechnicalAnalysis. The Assignment2 folder contains the report, the R script, portfolio configuration, README documentation, the CSV statistics output, and the Stock_Charts folder. The repository URL is:

https://github.com/mohamedelahwelca-hub/TechnicalAnalysis

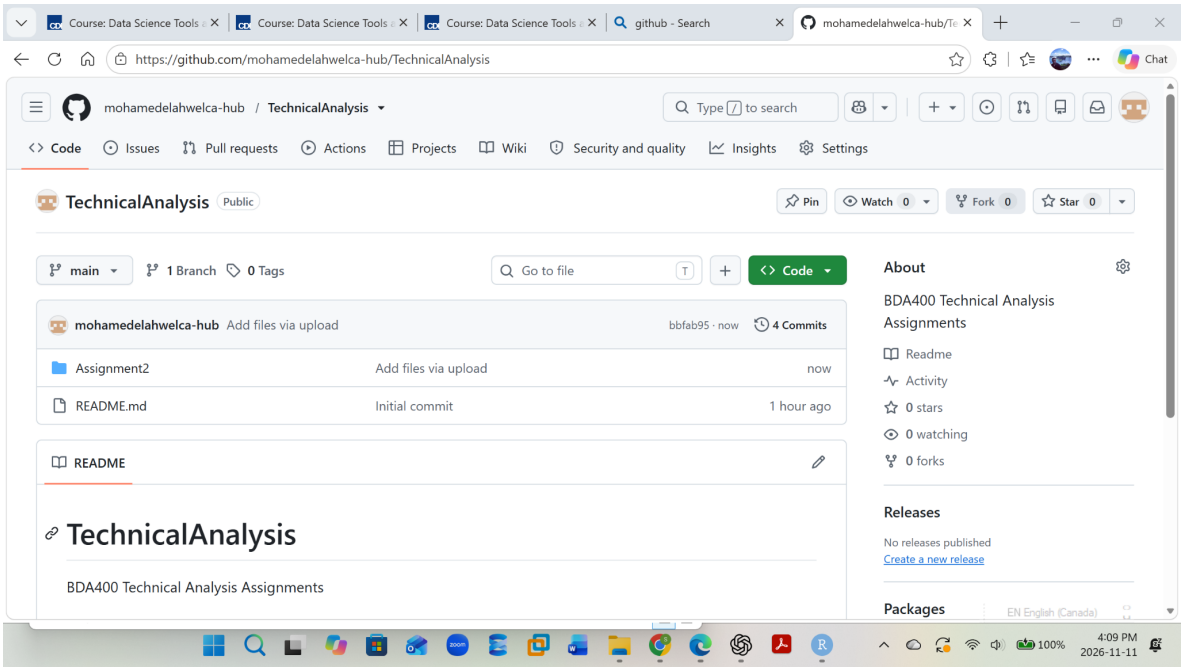


Figure 1. GitHub repository home page showing the TechnicalAnalysis repository and Assignment2 folder.

3. Assignment2 Folder Contents

The repository screenshot below shows the Assignment2 directory and the project deliverables. The folder includes both DOCX and PDF report versions, README files, portfolio.txt, technical_analysis.R, stock_statistics.csv, and the Stock_Charts directory.

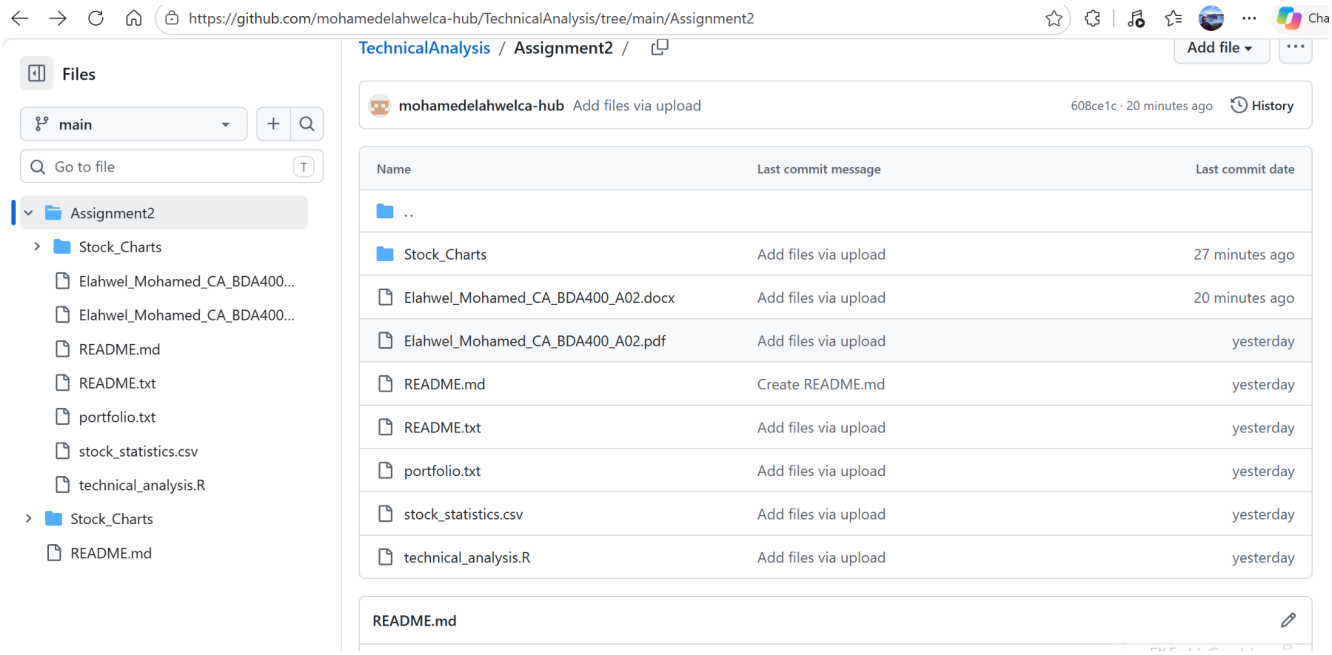


Figure 2. GitHub Assignment2 directory showing the uploaded project files.

4. RStudio Environment and Required Packages

The R script is designed to check for and load the packages quantmod, TTR, xts, and ggplot2. quantmod supports market-data retrieval, TTR provides technical-analysis calculations, xts handles time-indexed data, and ggplot2 is used to create charts. The RStudio screenshot confirms that the script was open and the analysis workflow had been run in RStudio.

The available evidence verifies an active RStudio session and successful analysis output. A separate screenshot of the original software installation is not included.

5. Data Import and Statistical Analysis

The project reads the ticker symbols from portfolio.txt and processes each available stock series. The analysis calculates the latest 20-day simple moving average, mean, mode, median, and standard deviation of closing prices. The following values are transcribed from the RStudio Console output shown in Figure 3.

Symbol	20-day MA	Mean	Mode	Median	Std. Dev.
AAPL	344.2325	293.6280	272.95	289.36	30.60415
MSFT	531.0945	448.7340	483.47	451.10	53.41159
NVDA	206.1830	200.3415	210.96	199.00	17.68746

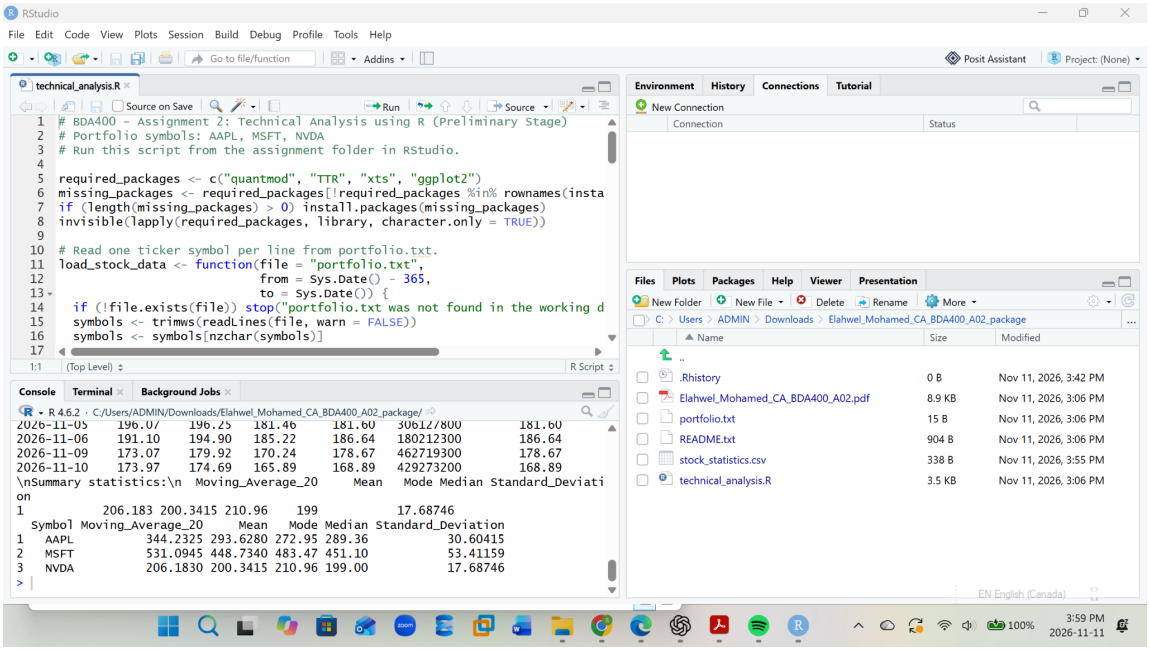


Figure 3. RStudio Console output showing recent stock data, summary statistics, and the generated CSV file in the Files pane.

6. Visualization and Output Files

The script generates a closing-price time-series chart for each ticker and saves the charts as PNG files in the Stock_Charts folder. The RStudio evidence below shows the NVDA Closing Price chart together with the statistics output. The project also creates stock_statistics.csv as a separate output file.

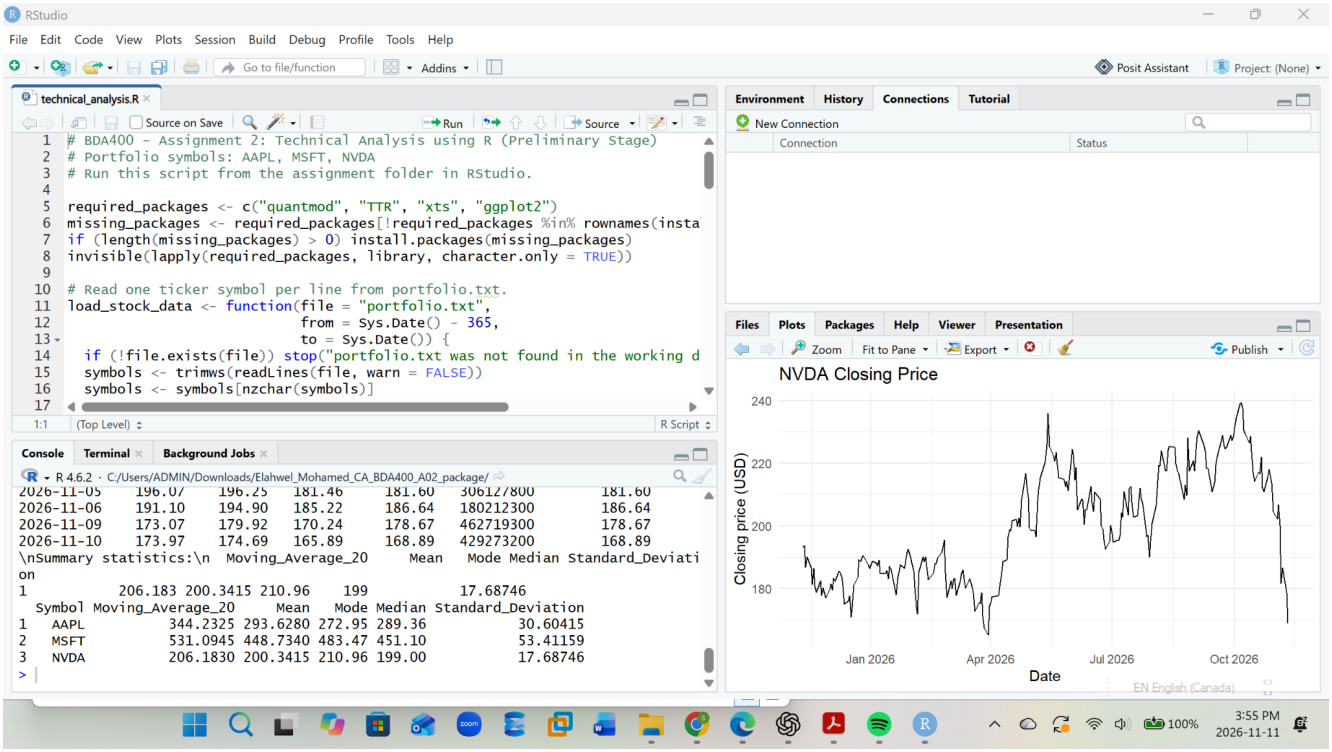


Figure 4. RStudio chart output for NVDA closing prices, with the summary statistics visible in the Console.

7. How to Run the Project

1. Open the Assignment2 folder as the RStudio working directory.
2. Confirm that portfolio.txt and technical_analysis.R are in that folder.
3. Open technical_analysis.R and run the script from top to bottom, or use Source.
4. Allow the script to install any missing required packages.
5. Review the Console for imported data and calculated statistics.
6. Review the Plots pane and the Stock_Charts folder for chart outputs.
7. Confirm that stock_statistics.csv is present, then commit and push updated files to GitHub.

8. Conclusion

The available screenshots demonstrate that the RStudio analysis was run, summary statistics were produced for AAPL, MSFT, and NVDA, a closing-price chart was displayed, and the project files were uploaded to the GitHub Assignment2 directory. This establishes the main preliminary workflow for the technical-analysis project.

9. Final Submission Checklist

Requirement	Evidence / Status
GitHub repository named TechnicalAnalysis	Repository screenshot included; URL provided
Assignment2 folder and project files	GitHub folder screenshot included
RStudio project execution	RStudio Console screenshot included
Stock symbols AAPL, MSFT, NVDA	Listed in report and analysis output
Summary statistics	Values transcribed into results table
Closing-price chart	RStudio chart screenshot included
CSV statistics output	stock_statistics.csv visible in RStudio Files pane
R and RStudio installation screenshots	Not included in the available evidence

Before submitting: if the instructor explicitly requires screenshots of the R and RStudio installation process or package installation, add those genuine screenshots separately. Do not treat the RStudio execution screenshots as proof of the original installation process.

Repository: https://github.com/mohamedelahwelca-hub/TechnicalAnalysis

Appendix A – RStudio Installation Evidence

The following genuine screenshots document the RStudio installation process on the student's computer. They show the setup welcome screen, selected installation location, installation progress, and final confirmation that RStudio was installed successfully.

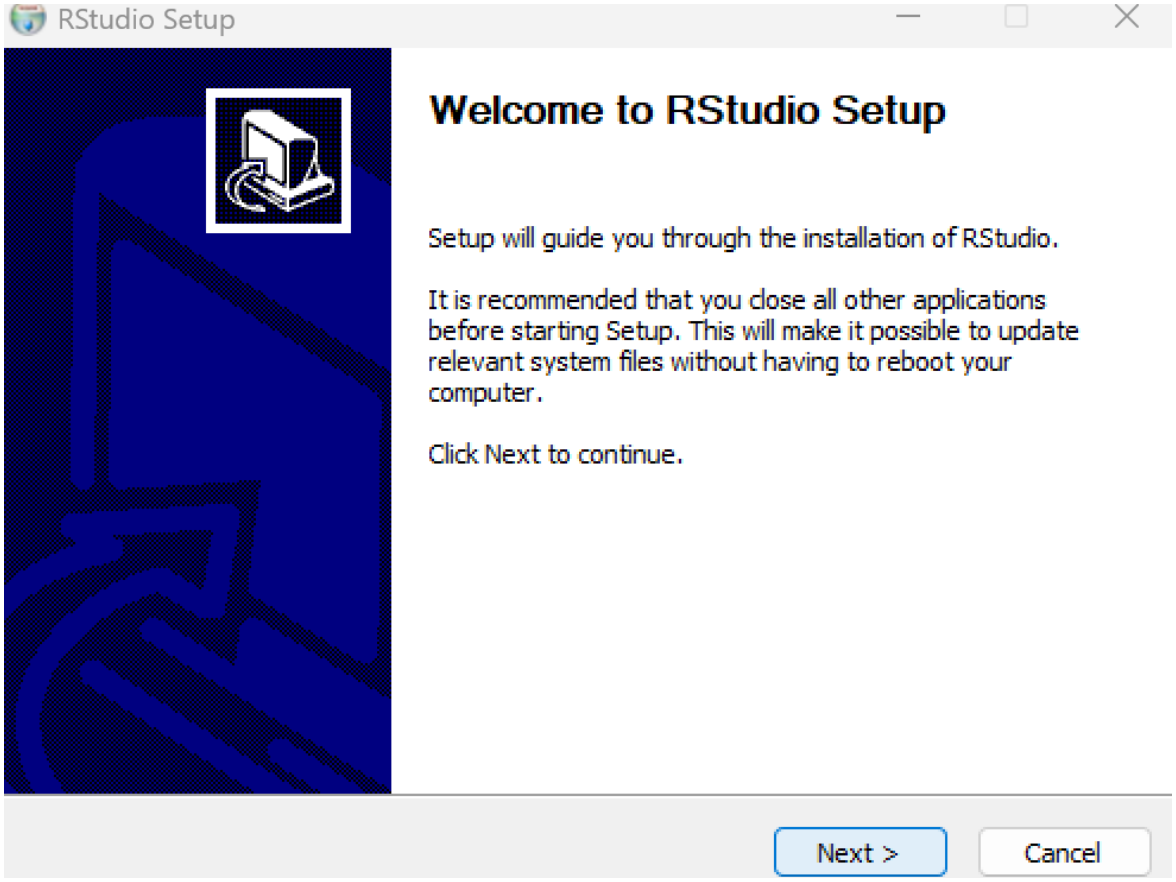


Figure 5. RStudio Setup welcome screen.

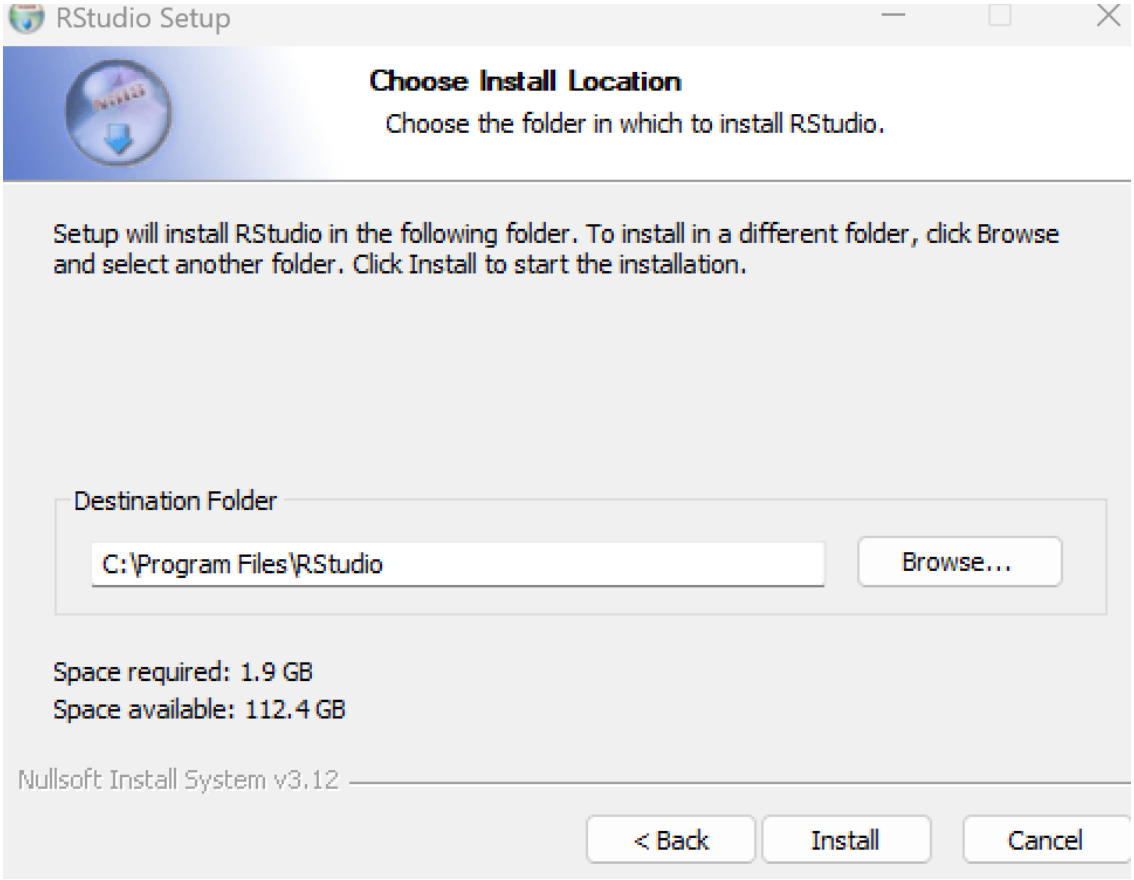


Figure 6. RStudio installation destination folder.

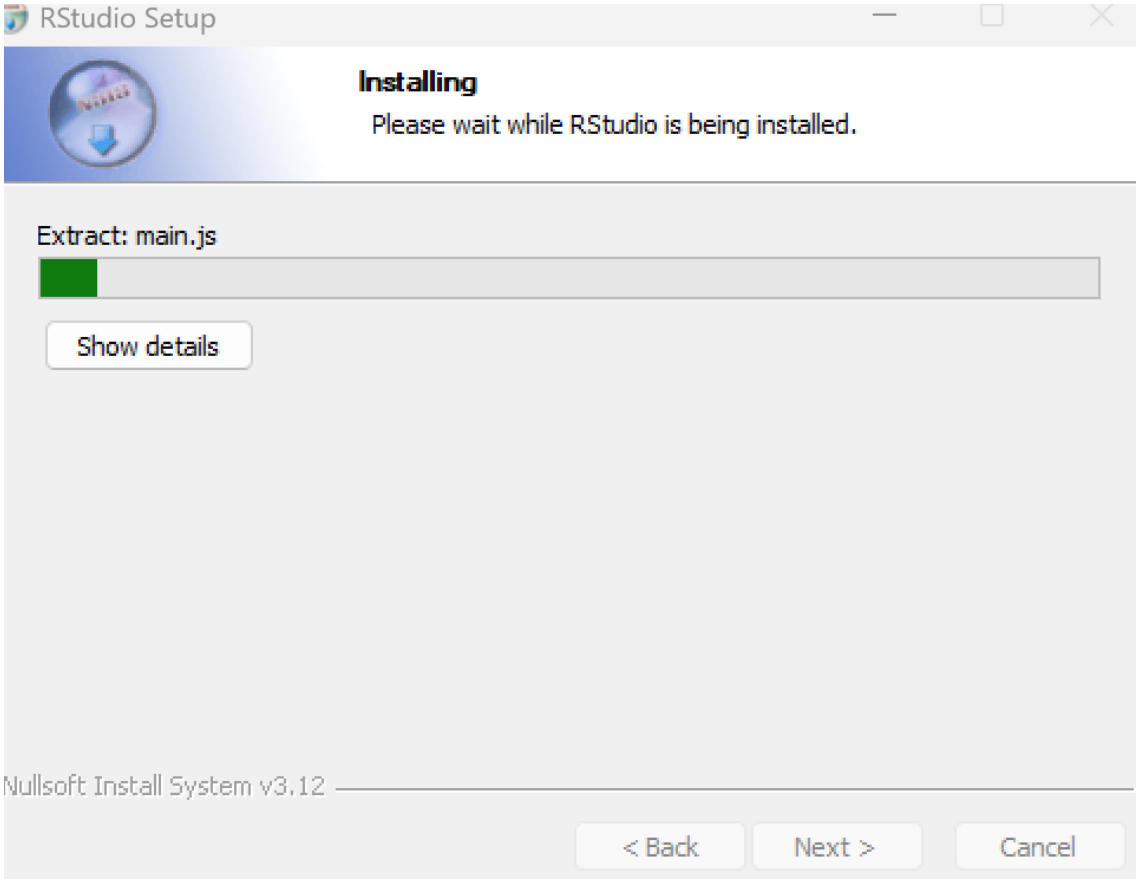


Figure 7. RStudio installation in progress.

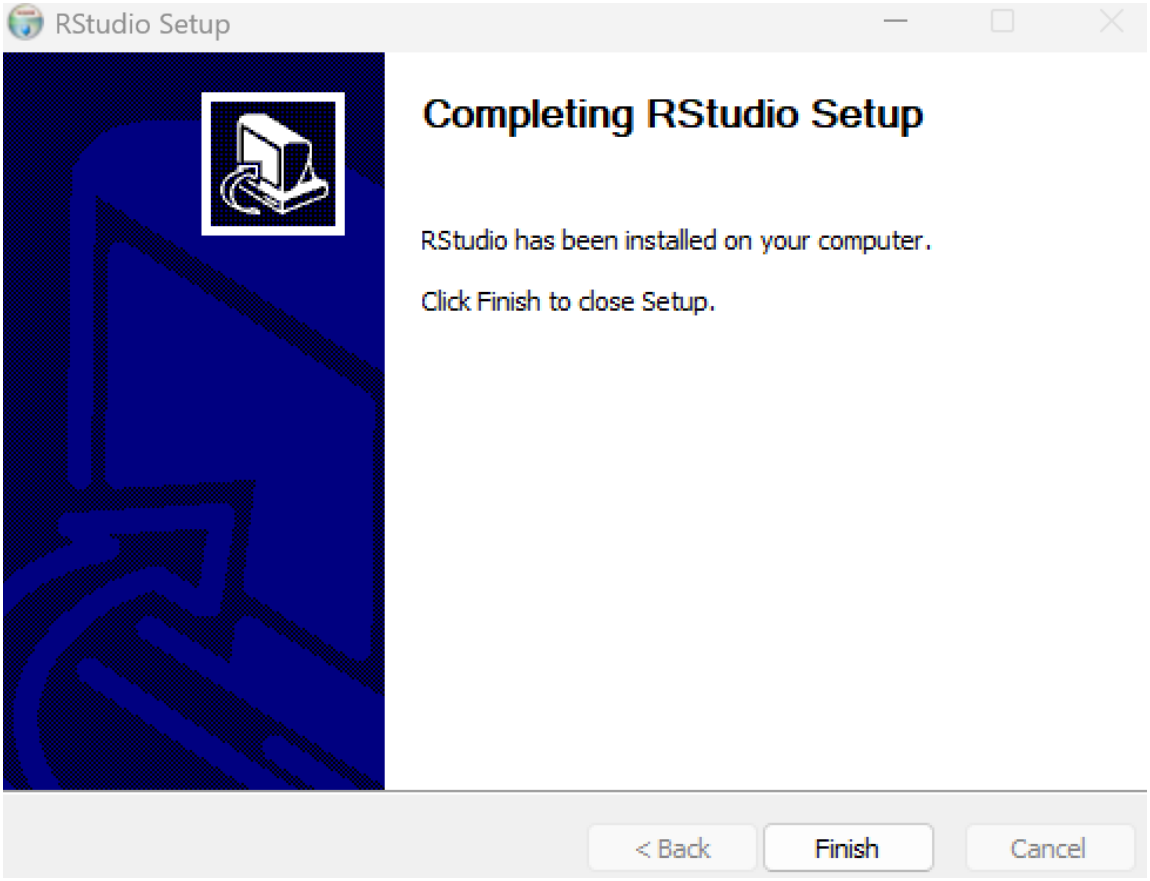


Figure 8. RStudio Setup completion screen confirming successful installation.